

Project & Team Overview

Project Title: CloudPulse — Multi-Cloud Cost Reclamation & Instant Hydration Engine

Team Name: ARGUS Innovators

Track / Domain: AI Infrastructure, Green Computing & Business Automation

Role	Team Member Name	Institution / Department
Team Leader	Lead Innovator	Thiagarajar College of Engineering / Partner Inst.
Core Engineer 1	Backend & AI Specialist	Department of Computer Science & AI
Core Engineer 2	Cloud & Hardware Specialist	Department of Electrical & Electronics

The \$17 Billion Cloud Waste Challenge

Massive Off-Hours Waste:

- Non-production environments (staging, dev, testing) run 24/7 but are active only ~40 hours/week.
- Over **120 off-hours weekly** generate pure financial waste across AWS, GCP, and Kubernetes.

Critical Flaws in Advisory FinOps Tools:

- **Advisory Friction:** Legacy tools send static PDF digests/tickets that developers ignore due to outage fears.
- **Single-Metric Flaws:** Scripts that check CPU utilization alone routinely corrupt active DB background jobs.
- **Re-Hydration Bottlenecks:** Restoring stopped servers manually takes 30-60 mins, crippling developer speed.

Proposed Solution & System Architecture

CloudPulse Core Solution: An autonomous multi-cloud execution engine combining AI telemetry with instant developer re-activation.

- **Multi-Signal Telemetry Fusion:** Fuses CPU, network throughput, and active DB/HTTP sockets.
- **Sub-3-Second Instant Hydration:** 1-click Web UI & Slack slash commands (`/cloudpulse wakeup``).
- **Ghost Resource Reaper:** Purges unattached storage volumes and orphan static Elastic IPs safely.

Next.js 14 Web Dashboard

(Re-Activation Portal / Ghost Sweeper / Carbon Offset Ledger)

■ REST API / OpenAPI Specification

FastAPI FinOps Core Engine

(Tag Discovery -> Multi-Signal Evaluator -> Execution Reaper)

■ Multi-Cloud SDKs & Edge Hardware Gateway

(AWS Boto3, GCP Compute API, Kubernetes Client, C-DAC VEGA RISC-V SoC)

Artificial Intelligence & Technology Stack

AI Models & Algorithmmic Innovation:

- **Multi-Signal Telemetry Evaluation:** Logical AND fusion across rolling 30-minute metric streams (CPU < 2.0%, Network < 10KB/s, Connections == 0).
- **Workload Usage Pattern Analytics:** Time-series analysis predicting off-hours developer activity to prevent unintended service interruptions.
- **Auditable Carbon Footprint Engine:** Calculates real-time CO2 emission offsets based on global grid energy intensity factors.

Architecture Layer	Technologies & Frameworks
Backend Engine	FastAPI (Python 3.11), Async SQLAlchemy, Pydantic, Boto3 SDK
Frontend Portal	Next.js 14 (App Router), React 18, Tailwind CSS, Recharts
Edge & Cloud Layer	AWS EC2/EBS/EIP, GCP Compute, Kubernetes, C-DAC VEGA RISC-V SoC Driver

Working Prototype & Engine Implementation

CloudPulse Core Modules Implemented & Functional:

1. **Discovery Engine** (`discovery.py`): Tag-aware resource discovery filtering out production workloads.
2. **Evaluator Engine** (`evaluator.py`): Multi-signal evaluator checking rolling CPU, network, and connection metrics.
3. **Executor Engine** (`executor.py`): Pauses EC2/GCE, scales K8s deployments to 0 replicas, purges orphan EBS/EIPs.
4. **Hydration Webhook** (`hooks.py`): Sub-3-second re-activation handler for 1-click Web UI & Slack ChatOps.
5. **Zero-Cloud Simulation Driver** (`simulated_driver.py`): Complete offline demo driver ready for live evaluation.

Innovation & Competitive Advantage

Benchmark Comparison against Legacy FinOps Solutions:

Evaluation Metric	Legacy Advisory FinOps	CloudPulse Autonomous Engine
Action Execution	Manual Tickets / Reports	100% Autonomous Execution
Production Safety	High Outage Risk (CPU-only)	0.0% False-Positive Outages
Re-Activation Speed	30 - 60 Minutes	< 2.8 Seconds Instant Warm
Ghost Asset Sweeping	Monthly Manual Audit	Real-Time Continuous Purge

Business, Social & Environmental Impact

Quantifiable Impact & Value Delivery:

- **Financial Cost Reclamation:** Up to **45% savings** on non-production cloud infrastructure billing.
- **Auditable Carbon Footprint Reduction:** Directly computes carbon emission offsets:
Carbon Offset (kg CO₂) = Total Idle Hours Saved × 0.20 kW × 0.385 kg CO₂/kWh
- **Developer Velocity Preservation:** Instant sub-3-second hydration eliminates developer waiting time.
- **Scalability:** Multi-cloud & container-native architecture supporting AWS, GCP, Azure, and Kubernetes.

Development Roadmap & Commercial Strategy

Current Progress:

- Core FastAPI Engine & Async Database Schema fully implemented.
- Multi-cloud discovery, ghost sweeping, and simulation drivers operational.
- Next.js 14 interactive dashboard with instant hydration hooks ready for live demo.

Future Commercialization Roadmap:

- **Phase 1 (Q4 2026):** Time-series predictive pre-hydration using developer git commit activity.
- **Phase 2 (Q1 2027):** Verifiable ESG Carbon Certificates on energy-efficient block ledgers (Web3).
- **Phase 3 (Q2 2027):** Enterprise SaaS product launch targeting mid-to-large tech companies.